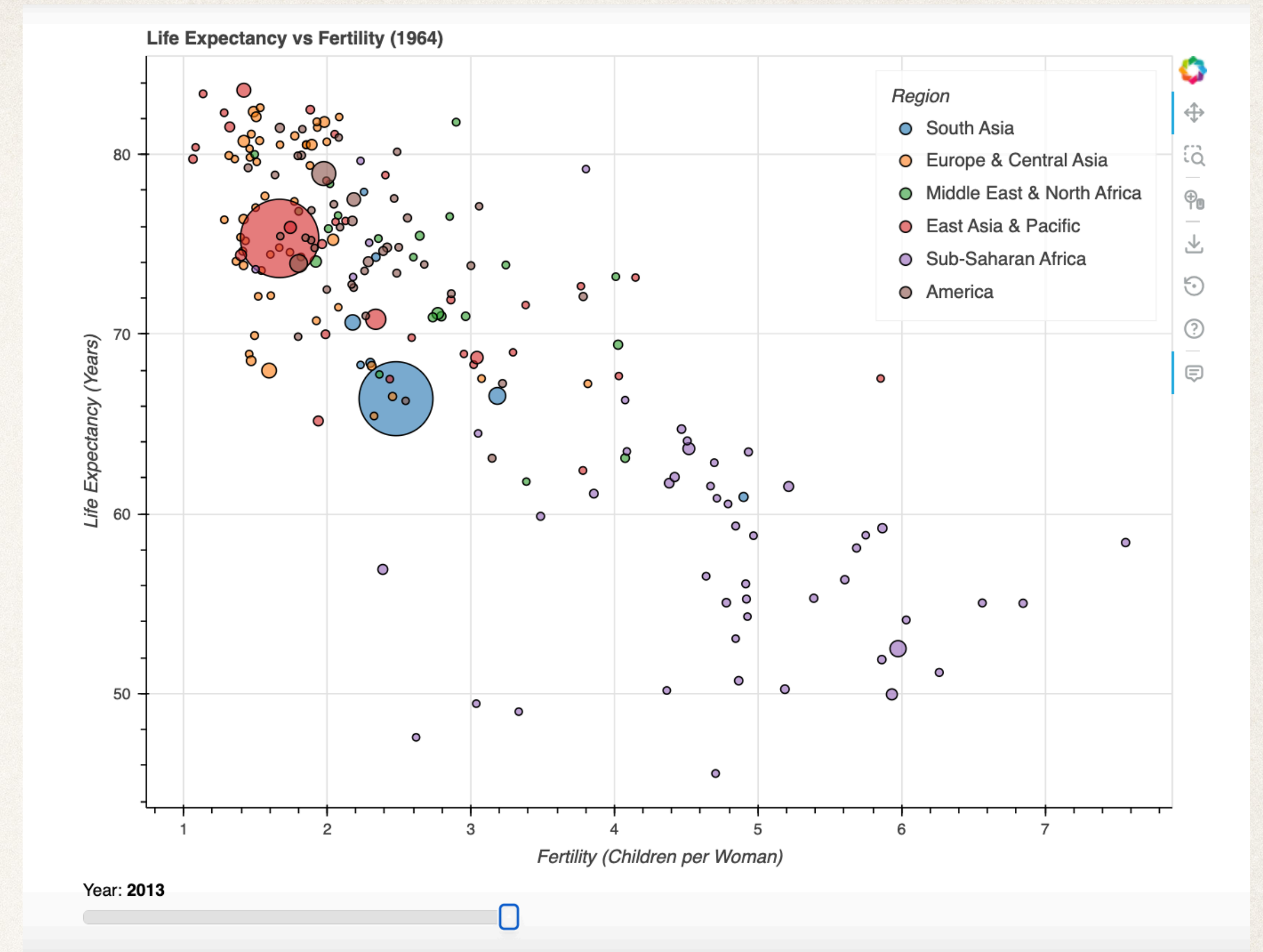
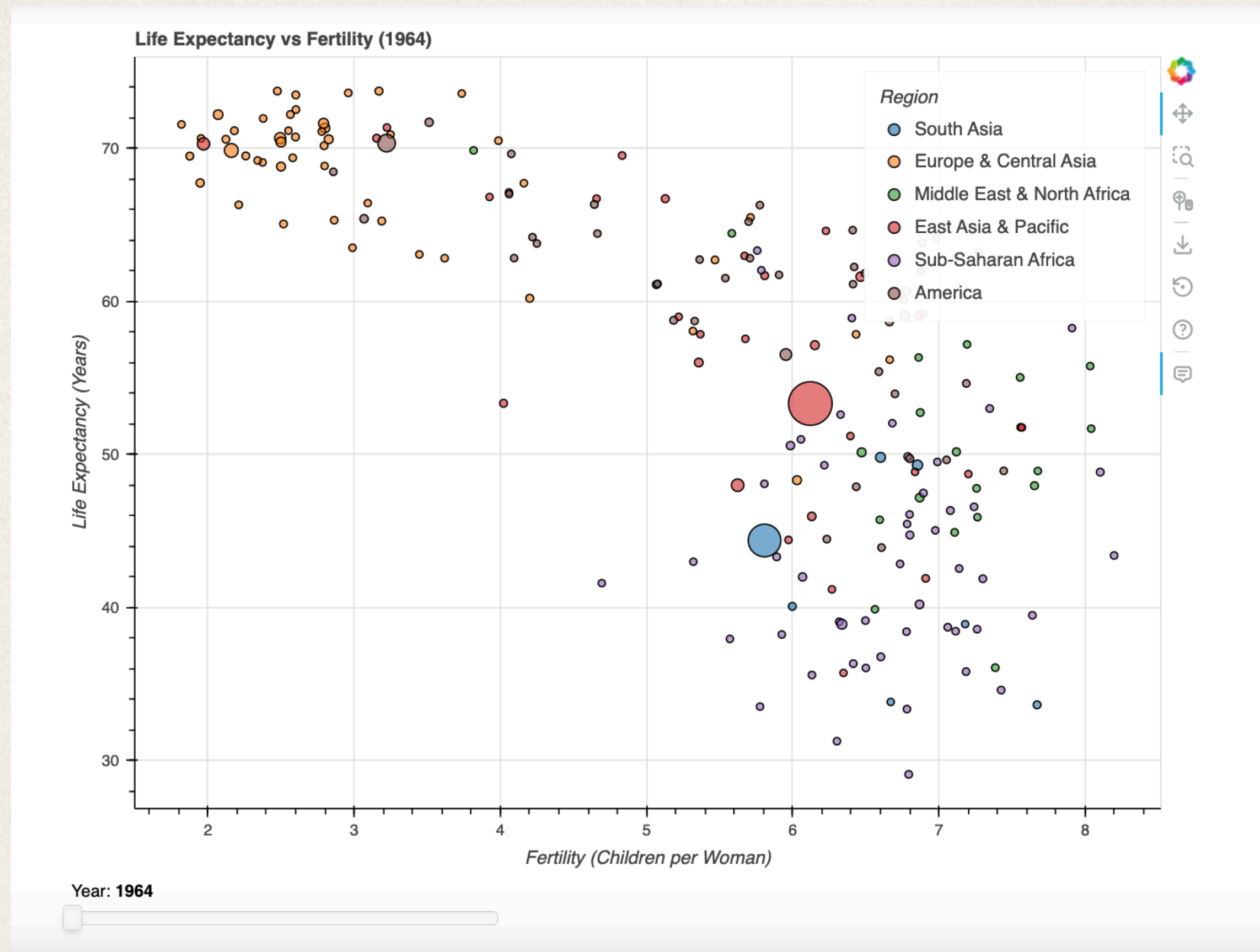




Life Expectancy and Fertility Visualization(1964-2013)

Interactive Data Visualization using Python and Bokeh ([chart of development trend](#))

Research Question

Main Question

How did life expectancy and fertility change between 1964 and 2013?

- Which regions developed the fastest?
- Did countries become more similar?

Goal

interactive visualization:

- compare regions
- long-term changes
- global development trends.

Dataset & Design Choice

Dataset

- Gapminder.org
- 244 countries
- 1964–2013
- CSV format

Variables

- Country
- Year
- Region
- Population
- Life expectancy
- Fertility

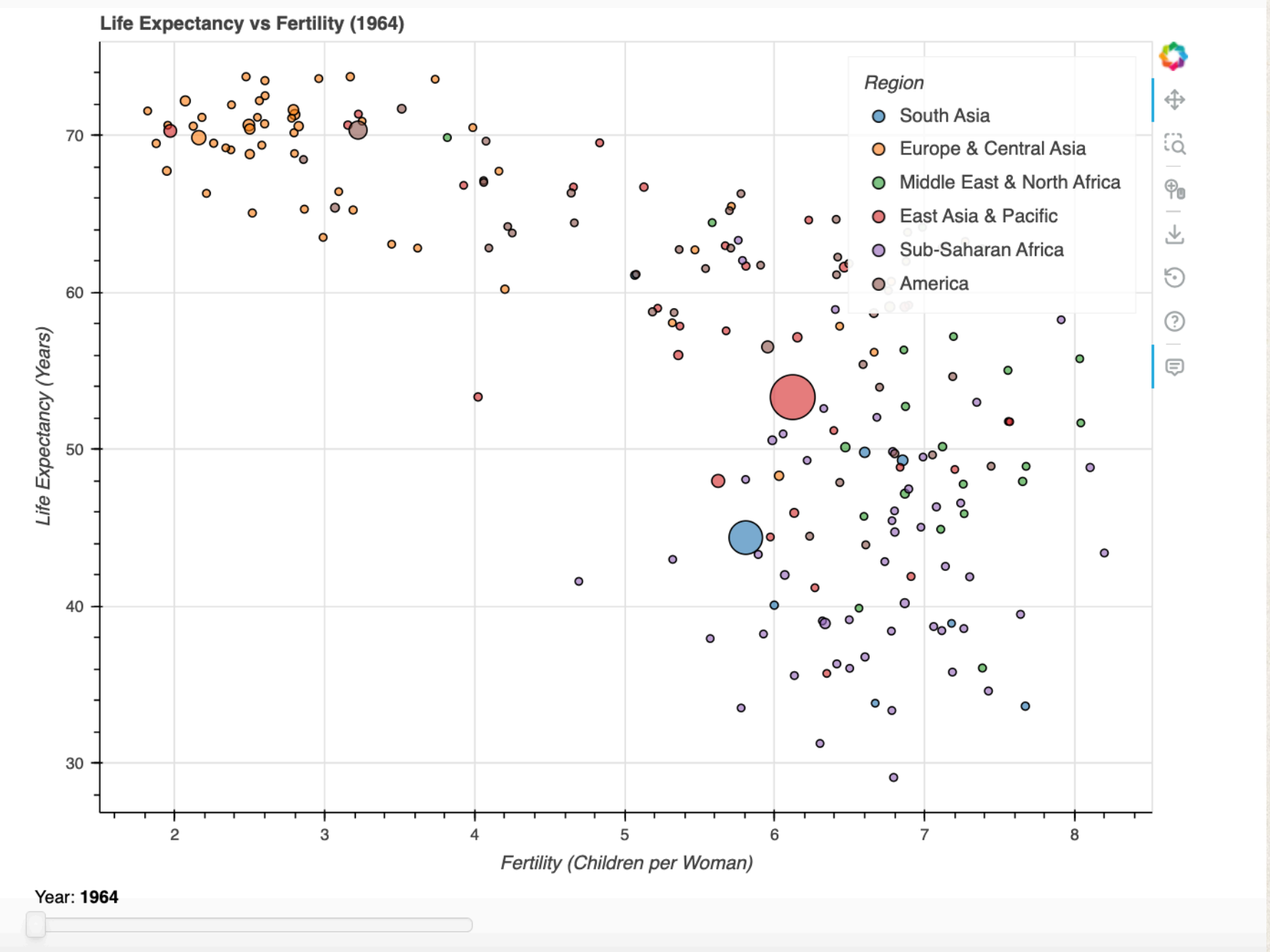
Scatter Plot Design

- X-axis → Fertility
- Y-axis → Life expectancy
- Bubble size → Population
- Color → Region

Purpose

- compare multiple variables
- regional differences
- time changes

```
p.scatter(  
    x='x',  
    y='y',  
    size='pop_size',  
    fill_color=palette[i]  
)
```



Interactive Features

Bokeh Features

- Year slider (change)
- Hover tooltips (mouse - details)
- Clickable legend (hide specific)
- Dynamic updates chart

Hover Information

- Year
- Country
- Population
- Fertility
- Life expectancy

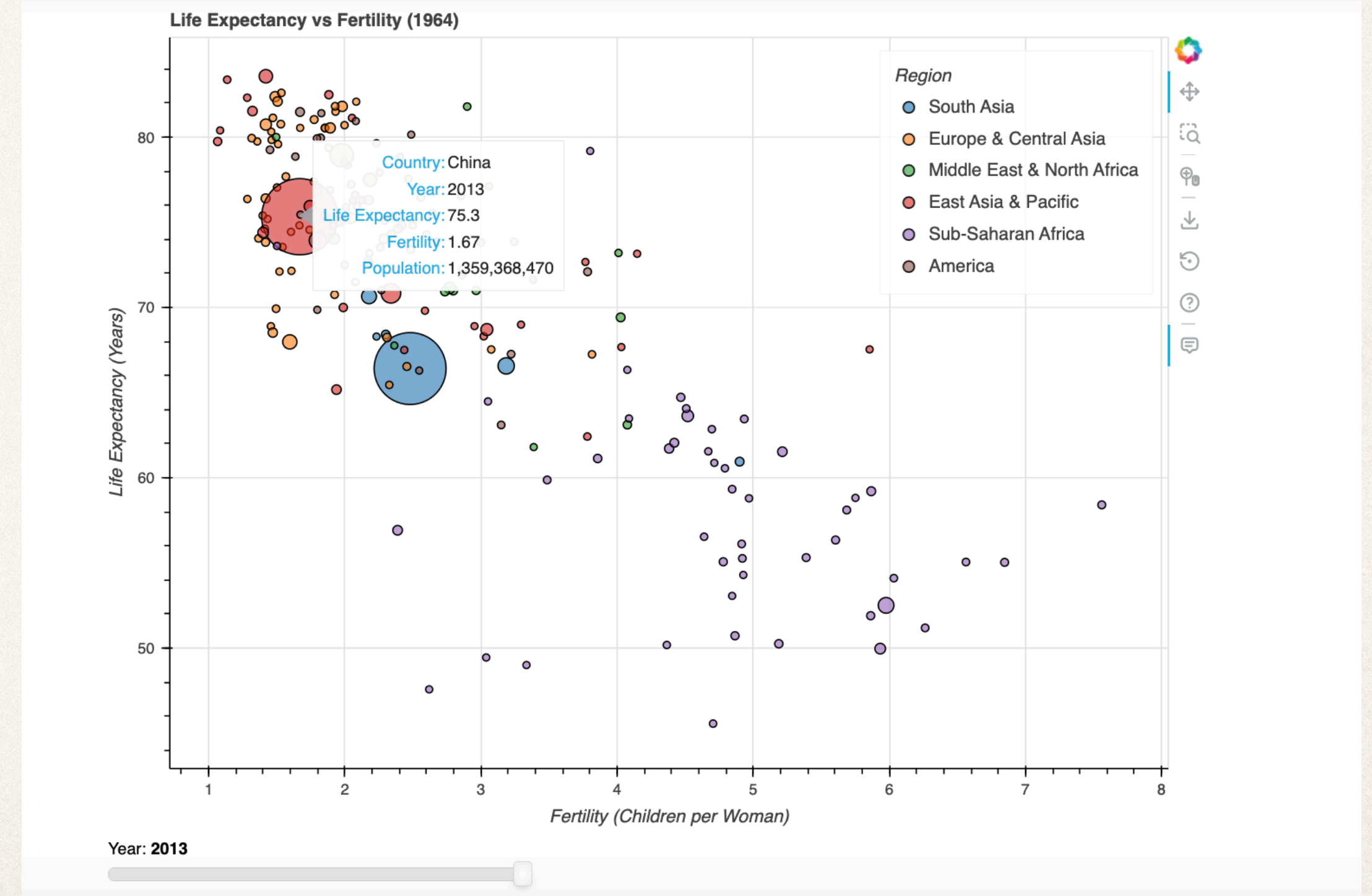
<https://rongmans.github.io/final/>

```
# HoverTool
hover = HoverTool(tooltips=[
    ("Country", "@country"),
    ("Year", "@year"),
    ("Life Expectancy", "@y{0.0}"),
    ("Fertility", "@x{0.00}"),
    ("Population", "@population{0,0}"),
])
p.add_tools(hover)

p.legend.title = "Region"
p.legend.location = "top_right"
p.legend.click_policy = "hide"

slider = Slider(start=years[0], end=years[-1], value=initial_year, step=1, title="Year")

# connects the slider with JavaScript callback, chart updates automatically when year changes.
slider.js_on_change('value', callback)
```



Workflow

Python → HTML → JavaScript → BokehJS

Data process

- ① Python/Pandas (CSV, scale pop, split region)
- ② Bokeh visualization (plot, slider, legend, hover)
- ③ User selects a year (slider Interaction)
- ④ JavaScript callback (filter data source)
- ⑤ BokehJS redraw (update chart in browser)

Main Learning

- Python + JavaScript + Bokeh
- interactive systems

⌕ JavaScript

```
if(data['Year'][j] === year &&  
    data['Region'][j] === region)
```

```
slider.js_on_change('value', callback)
```


Main Findings

Global Trends (chart shift)

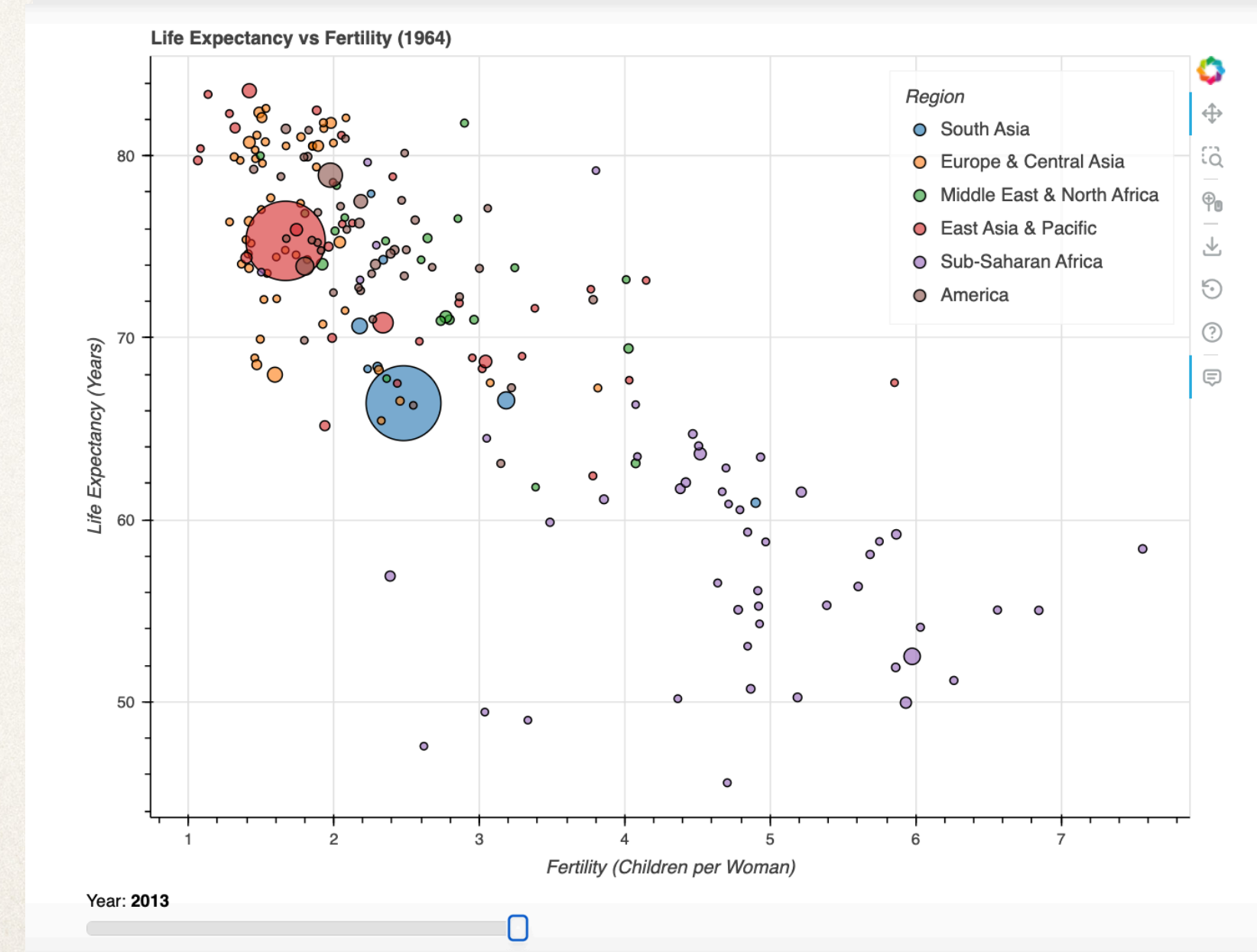
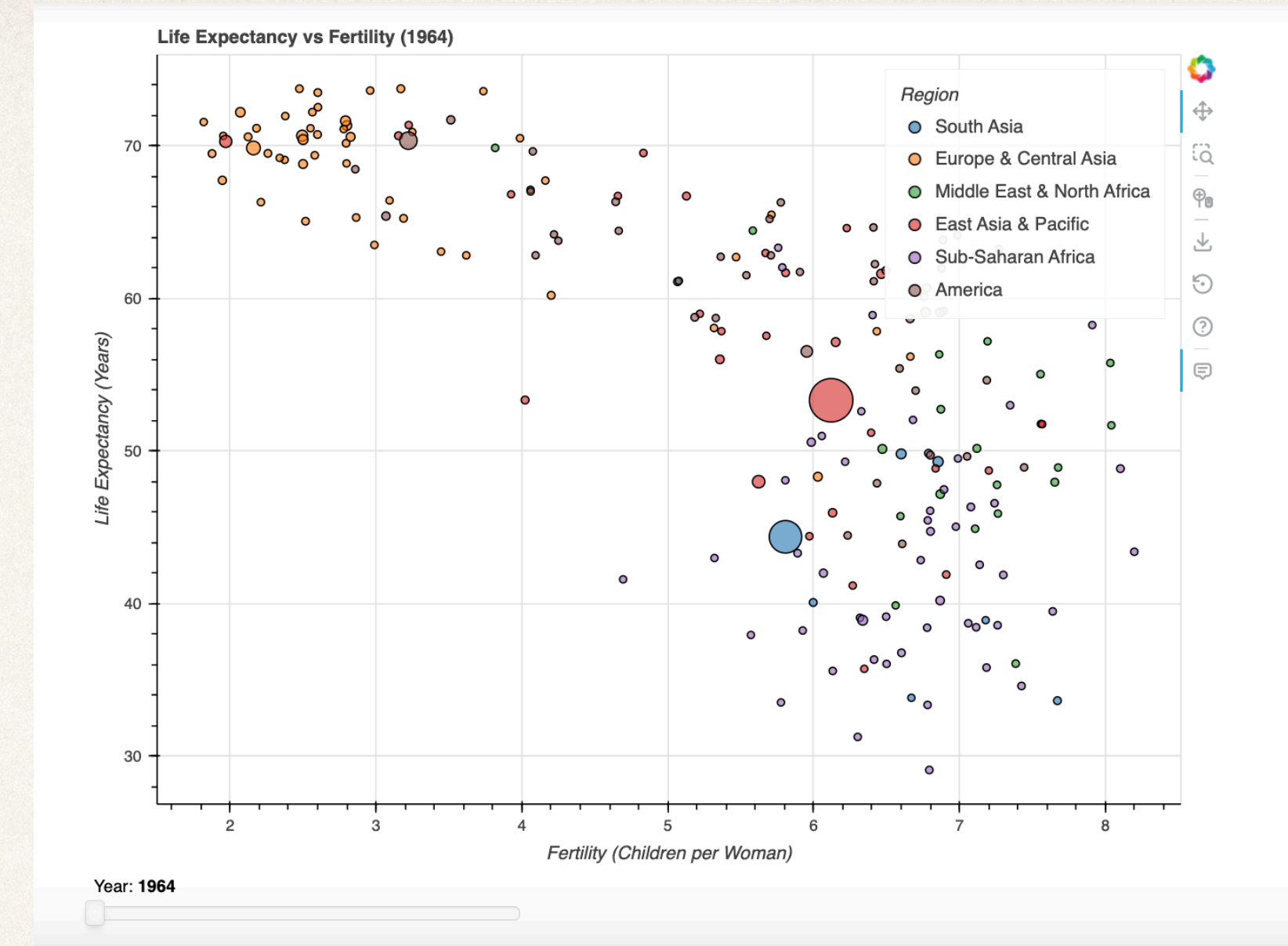
- Fertility  Life expectancy 

Regional Development (Life expectancy)

- Europe&North America developed earlier
- East Asia rapidly
- Sub-Saharan Africa slowly

Country Differences (points concentrated)

- Countries became similar



Limitations & Future Improvement

Limitations

- Overlapping countries
- Simplified population scaling
- Missing data
- Browser performance limits

Future Improvements

- Country search function
- Animation playback
- Map visualization

Challenges & Learning

Challenges

- First time using Bokeh
- JavaScript callbacks
- Multiple data sources

What I Learned

- Interactive visualization
- Data transformation (read + scale + split)
- Bokeh (Python + Javascript)
- GitHub (code + project)

design choices simplify data, interactive visualization better trends

Thanks